

Speaker Notes

Contactless Vital Sign Estimation from Face Video — A Rigor-First Audit of Reproducibility and Robustness in Remote Photoplethysmography (rPPG)

CSE754: Deep Learning for Computer Vision · Final Project Presentation

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1458 words total · approx. 10.4 minutes at a natural speaking pace

01 Contactless Vital Sign Estimation from Face Video

COVER 71 words · ~30s

Good [morning/afternoon]. This is our final project presentation for CSE754. Our project audits contactless heart-rate estimation, remote photoplethysmography or rPPG, for two things at once: does it stay accurate under different recording conditions, and can we trust the pipeline that produces that accuracy number in the first place. We'll walk through why this matters, the problem we targeted, our method, our results, one detailed failure case, and what we'd do next.

02 One Journey, Five Stops

AGENDA 128 words · ~55s

Five sections as required by the course, but one continuous journey: why this matters, the gap we found in the literature, how we built and validated our pipeline, what we measured once it was trustworthy, and where that leaves us. Since we covered the motivation and the literature gap in depth during our proposal presentation, we'll move through the first two sections briskly today as a refresher rather than repeating them in full, and spend most of our time from Method onward, where the actual hands-on work lives: the pipeline we built, the data we ran, and the results we measured. We'll keep coming back to a single idea throughout: rigor about the pipeline is not separate from rigor about fairness or robustness, it's a precondition for it.

03 A Camera Can Read Your Pulse, But Does It Read Everyone, Everywhere, Reliably?

01 · MOTIVATION 73 words · ~31s

Picture a telehealth visit with no pulse oximeter, just the webcam. Software silently estimates heart rate from color changes in skin too small to see. That already works in the lab. What's unresolved is whether it works consistently outside the lab, across conditions, and reliably enough to trust. Prior audits show error roughly tripling between easy and hard conditions, from about 5 bpm to over 14 bpm. That gap motivates everything that follows.

04 The Literature Gap: An Unaudited Intersection

02 · PROBLEM ADDRESSING 63 words · ~27s

Across the papers we reviewed, robustness work and fairness work never overlap on the same method. Even the most recent unsupervised state of the art has never been checked for consistency across conditions in a way that also questions whether the checking pipeline itself is trustworthy. That combination, robustness plus a rigor-first audit of the pipeline itself, is where we aimed our contribution.

05 Problem Statement and Research Scope

02 · PROBLEM ADDRESSING 80 words · ~34s

We want to be upfront about scope rather than let it surface as a surprise question. The original plan targeted skin-tone fairness using MMPD and EquiPleth, both gated-access datasets we could not obtain in time. Rather than claim an audit we couldn't actually run, we applied the same standard of rigor to resting-versus-post-exercise, the one real condition contrast our obtained data supports. The three questions on screen, tied together as one connected line of inquiry, are what we actually executed.

06 Pipeline Overview: A Six-Stage Evaluation Protocol

03 · METHOD 111 words · ~48s

Quick pacing note: we covered motivation and the literature gap in our proposal talk, so we'll move fast through those and spend most of today from here on: the actual pipeline, data, and results. Six steps, shown as one connected thread since each depends on the last. Steps 1 to 3 are infrastructure: versioned environment, preprocessing, and a subject-independent split, by person, never by frame, so no model can cheat by learning a face. Step 4 is the real gate: reproduction against published numbers. We don't touch steps 5 and 6, our actual evaluation and failure diagnosis, until step 4 genuinely passes. That's what the rest of today is built on.

07 Data and Baselines

03 · METHOD 101 words · ~43s

Twenty-two clips, pooled from both UBFC-rPPG releases: six from the original eight-subject set after excluding two for a duplicate-video issue, sixteen from the standard forty-two-subject benchmark the literature is measured against. We span both baseline families so our claims generalize: CHROM and POS, classical and training-free, and PhysNet under two pretrained checkpoints, one on UBFC-rPPG, one on PURE. Everything is inference-only; twenty-two clips is nowhere near enough to train from scratch without overfitting. Every number from here on is MAE and RMSE in bpm, computed per whole clip and always broken out by split, so it's directly comparable to the literature.

08 Validating the Pipeline: Implementation Defects Identified and Resolved

04 · OUTCOMES 99 words · ~42s

Each fix was verified by testing a rejected alternative, not accepted just because a number improved. The chart on the right is the strongest single piece of evidence: replacing the face detector alone cut error three to four times across every method. Worth flagging for anyone who's debugged a vision pipeline: none of these three bugs threw an exception or produced obviously broken output, they produced plausible numbers that were simply wrong. That's the failure mode that actually matters in this field, and it's why we treat implementation validation as inseparable from the science, not a formality before it.

09 Validated Baseline Reproduction

04 • OUTCOMES 74 words • ~32s

All four methods converge under 1.5 bpm error on the held-out test subjects. PhysNet, trained on an entirely different dataset and evaluated cross-dataset here, lands almost exactly on its published benchmark number, 1.62 against 1.63. That's not just close, it's within the kind of clip-level sampling noise you'd expect regardless, about as clean a reproduction as this field typically reports. This is the gate passing cleanly: the validated foundation everything after this builds on.

10 Diagnosing a Cross-Domain Accuracy Anomaly

04 • OUTCOMES 119 words • ~51s

This looked like an interesting anomaly at first, and we deliberately treated it as a signal to rule out a mundane bug before accepting an interesting explanation. We checked the checkpoint file wasn't swapped and checked our frame-rate handling, both clean. The real cause was a training-data mismatch, confirmed two independent ways: tracing source code and empirically re-testing on the correct domain, an eight-fold error reduction, 46.5 down to 5.8 bpm. The lesson generalizes well beyond our project: as more of the field adopts released pretrained checkpoints instead of training from scratch, a checkpoint's name is not proof of its training domain, and that exact gap is where a confident, wrong fairness or robustness claim would slip through unnoticed.

11 Cross-Condition Robustness: Resting versus Post-Exercise

04 • OUTCOMES 99 words • ~42s

Left: classical methods show a small but genuine gap between resting and post-exercise conditions, roughly three to seven bpm, post-exercise is modestly harder. Right, and more interesting: plot every method's per-subject error, and every method's worst failures land on a different subject. No subject is uniformly hard. Worth pausing on: a lot of robustness literature implicitly assumes some subjects are just inherently harder cases. Our data doesn't support that here; the residual error looks like each method's own instability, not a property of the person recorded, a distinction we lean on directly in the failure case coming up next.

12 Failure Case Analysis: Spectral Peak Diagnosis

FAILURE & INSIGHT 115 words • ~49s

Our largest single error, 72 bpm off. Before assuming the model failed to extract any pulse at all, we inspected the full spectrum, not just the winning peak. The true heart rate is right there: a strong second-place candidate at 71% of the winning peak's power. This reframes the failure as a peak-selection problem, not a signal-extraction failure, and it's a familiar failure mode to anyone who's worked with FFT-based peak picking on noisy, multi-modal spectra: argmax is a brittle decision rule exactly when two candidates are close, which is precisely the regime where it matters most. Signal present versus signal picked wrong is what actually determines whether a fix is worth building at all.

13 Mitigation Testing: A Rejected Candidate Fix

FAILURE & INSIGHT 108 words • ~46s

This is the second post-hoc fix we tested and rejected on the same standard: evaluate against the whole dataset, not the one case that motivated the idea. Both times, the fix traded one set of failures for a different set. That pattern is worth naming: it's the same failure mode as overfitting to a single validation example, just done by hand instead of by an optimizer, and it's exactly why we hold post-hoc heuristics to full-corpus evaluation rather than declaring victory on the motivating case. The real insight, and the actual lever that worked in this project: fixing the face detector, not tuning the estimator after the fact.

14 Summary of Contributions

05 · CONCLUDING REMARKS 105 words · ~45s

Four things we'd want you to remember, laid out as one connected line because that's really what they are: a single pipeline, built and stress-tested end to end. A working, validated pipeline. Three real bugs, each one individually capable of faking a fairness finding. An honest, appropriately caveated robustness measurement. And two negative results we chose to report rather than quietly drop. If there's one sentence to take away, it's the one on screen: rigor about the pipeline is a precondition for rigor about fairness. And if there's one habit worth taking into your own work: validate the harness before you trust what it measures.

15 Limitations and Future Directions

05 · LIMITATIONS & NEXT STEPS 79 words · ~34s

Being direct about scope: our test set is small, we never tested the skin-tone axis this was originally framed around, and one regression on the older dataset is still open. None of that undermines what we did establish, but we'd rather name the boundary clearly than let it surface as a surprise question. Each limitation here maps directly to a next step: the clearest one is going back to the original skin-tone ambition with this now-validated pipeline underneath it.

16 Thank You

CLOSING 33 words · ~14s

Thank you, we're happy to take questions. We have the full per-clip results, every figure, and both the classical and deep-learning code ready if you'd like to go deeper on any specific result.